T3000

Overview · built Nov 13, 2026

Overview

SOFTWARE

The T3 controllers have inputs, outputs, schedules and other features baked into the firmware which allow you to set up a typical control system quicklys. Here is a run down of the software feature list for the T3 controllers:

.

16 Control Basic programs of up to 10k total size

Each controller can be expanded with external T3 modules, up to a maximum of 64 inputs and 64 outputs per controller

128 Variables , these are used as virtual points in your calculations, displays and trend logs.

16 PID Controllers, used to control a process to a setpoint.

32 Trend logs, use these to see the temperatures and state of the building over extended periods.

Runtime totalizers and event logs

16 User screens [TBD: Expand to 128], used for floorplans and system diagrams.

8 Weekly schedules [TBD: expand to 16], used to schedule occupied periods.

4 Annual schedules [TBD: expand to 8], used to override occupied periods on holidays.

16 Variable arrays [TBD: add arrays]

8 User name/ password combinations [TBD: expand to 50]

5 Custom tables , used for custom sensors

8 Custom digital engineering units , on/off is standard but you can add custom states like purge/fill for example.

Digital/Analog alarms , used to alert the operator about failed equipment.

Programming editor

Graphical user screen editor for custom displays

JPG, BMP, PNG file format for floor plans, mechanical system diagrams and icons

T3000 OPERATING SYSTEM SOFTWARE

The T3000 software is used by both casual users for day to day interaction with the building as well as experienced operators and contractors when they are setting up a system. The software runs on any PC running Windows 7,8 &10 and there are no licensing restrictions allowing you to copy the software to as many machines as you like. There are no linux or Mac versions. Third party software can always be used with T3 controller hardware by integrating over Modbus and Bacnet protocols, but there will always be a few procedures that are best accomplished using the T3000 software.

GETTING AROUND THE SYSTEM

Most of the system configuration is done in a series of tables which the user can navigate and edit like working in a spreadsheet. The tables are updated dynamically with real-time information even as items are being edited. Navigate in the program using the menu system and mouse. There are hotkeys for many operations so experts can get around the system quickly, see the next section on the Menu and work areas for more info.

[TO BE DONE] There is a command line which is brought up by pressing the space bar and allows you to jump to 1INS for example, to quickly view the inputs on panel 1.

ALARMS

Alarms are generated by the T3 controllers based on readings and status of the field sensors. When a certain temperature is too hot for a certain period for example, the alarm will be generated. These alarms are programmed in the Control Basic language using the ALARM and DALARM statements. These statements allow the user to define their own logic and timing of the alarm statements which can be as complex or simple as you like. A typical simple alarm for the room temperature is shown here, if the room is over or under setpoint by 10 Degrees for more than 100 seconds the alarm will trigger.

200 DALARM , TEMP101 - SETPT101 > 10 , 100, ROOM101 IS VERY HOT

210 DALARM , TEMP101 - SETPT101 < -10 , 100, ROOM101 IS VERY COLD

When an alarm occurs, an alarm symbol is displayed at the bottom right of the screen. Clicking on this symbol brings up the alarm log where the user can choose to acknowledge or clear the alarm. The alarm message is logged to the microSD disk locally as well as on the hard disk of any PC connected to the system. Each alarm shows the panel number from which the alarm originated, the date, time and alarm priority.

More details on the alarm statements and alarm handling are described here.

GRAPHICAL INTERFACE

The T3000 operating system can display floor plans and mechanical diagrams of the building using bitmaps which can be generated using any popular graphics package. Use a program such as Autocad, Corel Draw or Visio to prepare the graphic templates of the building and systems. These jpegs are stored on the hard disk of the PC and are used in the user interface of the particular building. Real time data is overlaid on the picture to show the temperatures, setpoints and so on of the system. You can move elements around on the display, add items and reconfigure the system interactively as the building operator works in the system. The screens can be nested to several levels deep enabling an operator to navigate through the entire building via the linked pages. Icons pointing to the trend logs, programming, and any associated item can be placed on the screen, there are a few canned icons and the user can create their own custom icons.

1.8.4 CONTROL BASIC

Control Basic is the programming language used for the logic running in the T3 controllers. There are many new modern languages which can be considered superior for their particular tasks but when it comes to programming a building, control basic gets the job done and its easy to learn and maintain later on. A sample of the programming goes like this:

10 IF SCHEDULE THEN START FAN101

20 IF TEMP1 < SETPOINT1 THEN START HEAT1

30 IF TEMP1 > SETPOINT1 + 1 THEN STOP HEAT1

Looking at these three lines you can see the programming is easy to read, the usual 'sequence of operations' from the mechanical system designer can be translated into basic programming quickly using the many built in control basic functions. The programming is stored in the local flash memory of the controller and starts running as soon as power is applied to the panel. One special feature of the T3 system is the ability to tweak the program while the building is running, there's no need to jump between software tools or take the panel offline. Just edit the program and hit the send key, the program is sent to the controller and runs immediately.